

Thermodynamics and ATP synthesis(Part 02)

Quantitative answers for the following questions



1. Will this happen spontaneously ? (Driving Force)
2. How much will this happen ?
3. Relation between 1 and 2 ?
4. How all this is related to biology ?
5. ~~How fast will this happen ? (KINETICS)~~

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2. How much will this happen ?

3. Relation between 1(driving force) &2 ?

➤ **Quantitative relationship : Free energy change and concentration of reactant and product.**

Plan of action: Establish the relationship for a simple case of ideal gases and by analogy for reactions in liquid

Quantitative relationship : Free energy change and concentration of reactant and product.

$$G = H - TS$$

$$\Rightarrow dG = dH - TdS - SdT$$

By definition $H = U + PV$?

$$\Rightarrow dG = (dU + PdV + VdP) - TdS - SdT$$

1st Law $dU = Q + W = TdS - PdV$ (For reversible process, No Non-Mech work)

$$\Rightarrow dG = TdS - PdV + PdV + VdP - TdS - SdT$$

$$\Rightarrow dG = VdP - SdT$$

At constant Temp

$$\Rightarrow dG = VdP$$

For 1 mole ideal gas, $PV = RT$

$$\Rightarrow dG = \frac{RT}{P} dP$$

$$\Rightarrow dG = \frac{RT}{P} dP$$

Integrate from P^0 (reference state 1 atm) to P

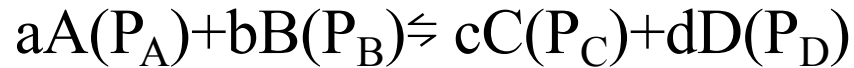
$$\Rightarrow G - G^0 = RT \ln(P/P^0)$$

Lets return to a prototype reaction : $aA(P_A) + bB(P_B) \rightleftharpoons cC(P_C) + dD(P_D)$

$$\Delta G = (G_{product} - G_{reactant}) = c G_C + d G_D - a G_A - b G_B$$

Substituting $G - G^0 = RT \ln(P/P^0)$

$$\Rightarrow \Delta G = (c G_C^0 + d G_D^0 - a G_A^0 - b G_B^0) + RT \ln \left[\frac{\left(\frac{P_C}{P^0}\right)^c \left(\frac{P_D}{P^0}\right)^d}{\left(\frac{P_A}{P^0}\right)^a \left(\frac{P_B}{P^0}\right)^b} \right]$$



$$\Rightarrow \Delta G = (c G^0_C + d G^0_D - a G^0_A - b G^0_B) + RT \ln \left[\frac{\left(\frac{P_C}{P^0}\right)^c \left(\frac{P_D}{P^0}\right)^d}{\left(\frac{P_A}{P^0}\right)^a \left(\frac{P_B}{P^0}\right)^b} \right]$$

$$\Rightarrow \Delta G = \Delta G^0 + RT \ln \left[\frac{\left(\frac{P_C}{P^0}\right)^c \left(\frac{P_D}{P^0}\right)^d}{\left(\frac{P_A}{P^0}\right)^a \left(\frac{P_B}{P^0}\right)^b} \right]$$

At equilibrium

$$\Rightarrow 0 = \Delta G^0 + RT \ln \left[\frac{\left(\frac{P_C}{P^0}\right)^c \left(\frac{P_D}{P^0}\right)^d}{\left(\frac{P_A}{P^0}\right)^a \left(\frac{P_B}{P^0}\right)^b} \right]_{\text{eqm}}$$

$$\Rightarrow \Delta G^0 = -RT \ln \left[\frac{\left(\frac{P_C}{P^0}\right)^c \left(\frac{P_D}{P^0}\right)^d}{\left(\frac{P_A}{P^0}\right)^a \left(\frac{P_B}{P^0}\right)^b} \right]_{\text{eqm}}$$

$$\Rightarrow \Delta G^0 = -RT \ln K_{\text{eqm}}$$

K_{eqm} = Equilibrium constant

ΔG^0 = Constant =

Standard free energy representing a hypothetical reaction with all of reactants and products at a pressure of 1 atm.

$$P^0_{(A/B/C/D)} = 1 \text{ atm}$$

Biological reactions do not occur in the gas phase. What about the free energy in solution. Conceptually no difference. Replace pressure by concentrations.

$$G - G^0 = RT \ln(\text{conc}/\text{conc}^0)$$

$$C^0 = 1 \text{ mol/lit}$$

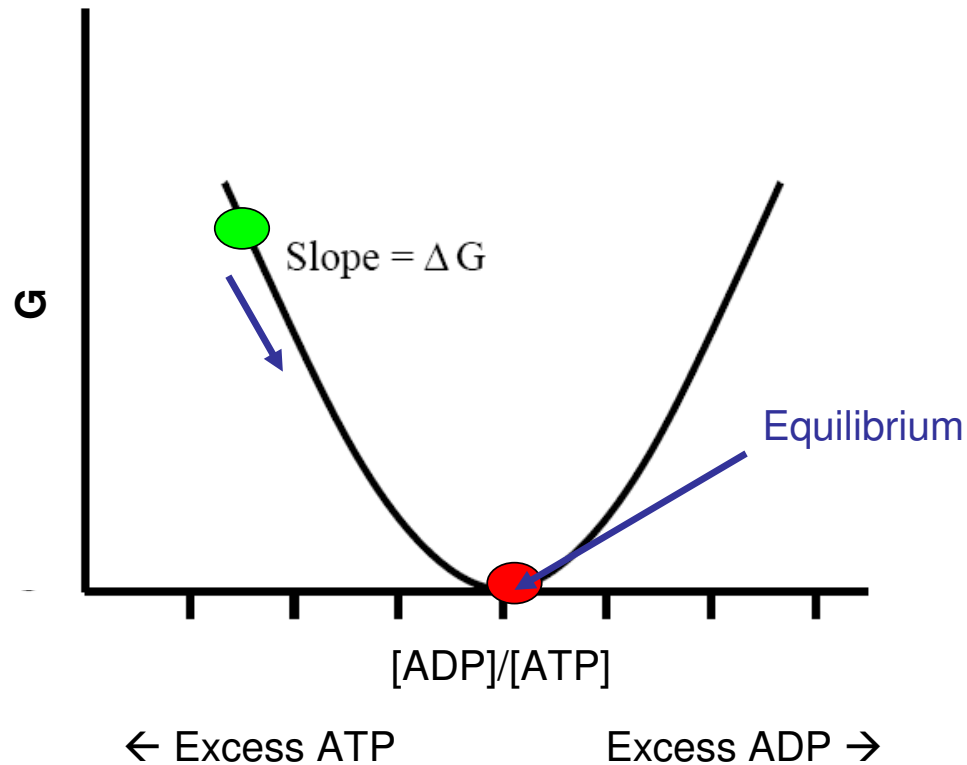
NOT IN EQUILIBRIUM

$$\Delta G = \Delta G^0 + RT \ln \left[\frac{\left(\frac{C_C}{C_C^0}\right)^c \left(\frac{C_D}{C_D^0}\right)^d}{\left(\frac{C_A}{C_A^0}\right)^a \left(\frac{C_B}{C_B^0}\right)^b} \right]$$

AT EQUILIBRIUM ($\Delta G = 0$)

$$\Delta G^0 = - RT \ln \left[\frac{\left(\frac{C_C}{C_C^0}\right)^c \left(\frac{C_D}{C_D^0}\right)^d}{\left(\frac{C_A}{C_A^0}\right)^a \left(\frac{C_B}{C_B^0}\right)^b} \right]_{\text{eqm}}$$

System Away from Equilibrium



For ATP hydrolysis to ADP

$$K'_{eq} = 2 \times 10^5 \text{ M},$$

For constant $[P_i] = 0.01 \text{ M}$

$$[ADP]_{eq}/[ATP]_{eq} = 2 \times 10^5 / 0.01 = 2 \times 10^7 \text{ M}$$

In the example in last slide:

$$[ADP]/[ATP] = 10^{-3}$$

Displacement from equilibrium gives the cell its ability to do work!

Energetics of ATP



Concentration of cellular Pi (remains almost constant), $[\text{Pi}] = 10^{-2} \text{ M}$

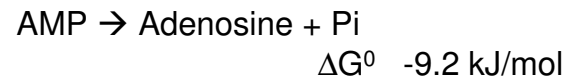
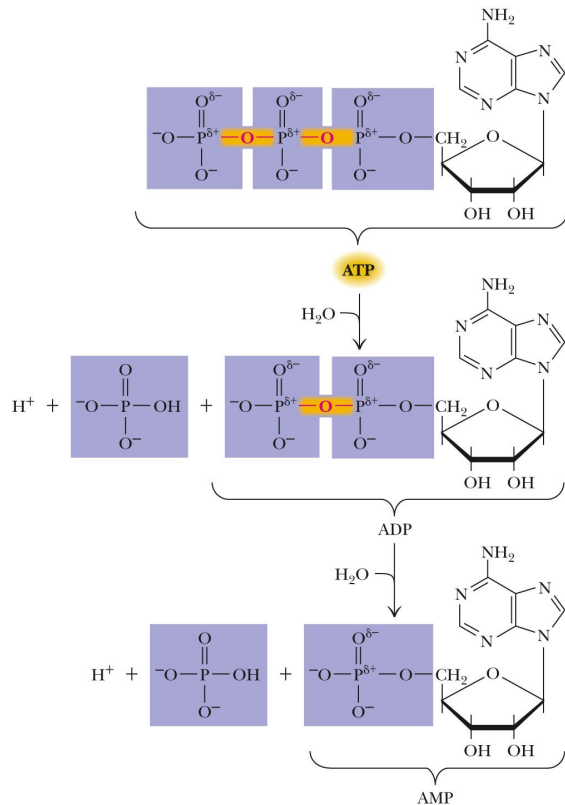
Concentration of ATP, $[\text{ATP}] = 10^{-2} \text{ M}$

Concentration of ADP, $[\text{ADP}] = 10^{-5} \text{ M}$

$$R = 8.3 \text{ J.mol}^{-1}.\text{K}^{-1}; T = 310 \text{ K}$$

$$\begin{aligned} \Delta G &= \Delta G^0 + RT \ln ([\text{ADP}][\text{Pi}]/[\text{ATP}]) \\ &= -30.5 + 8.3 \cdot 310 \ln(10^{-5}) = -60 \text{ kJ/mol} \end{aligned}$$

Energetics of ATP



Factors in the large negative ΔG^0 :

- electrostatic repulsion.
- stabilization of products by resonance.

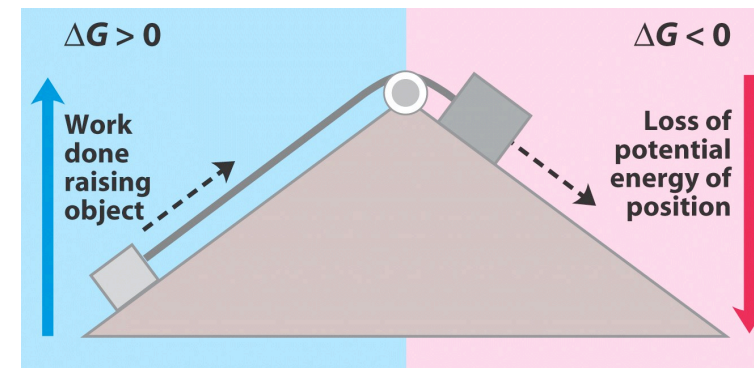
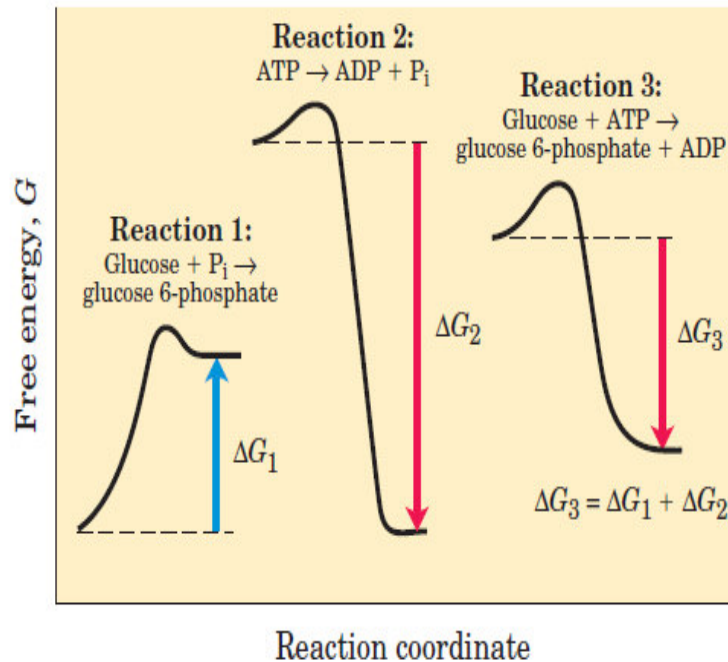
Energy Coupling:



Coupled reaction:



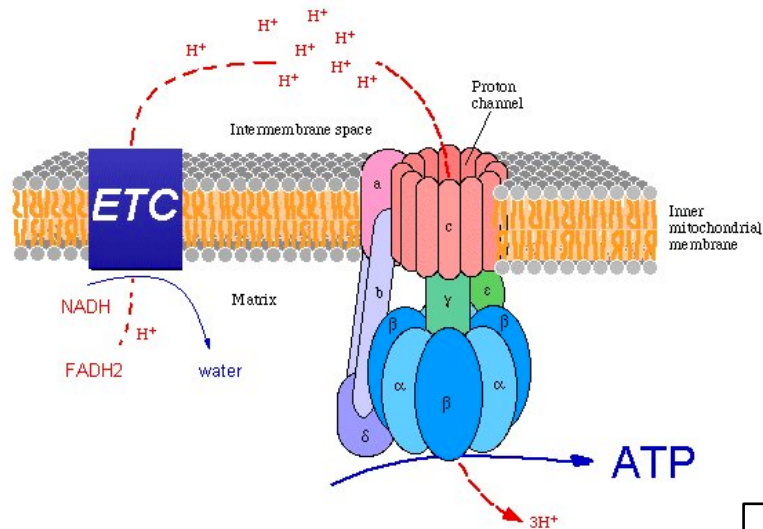
Half-reactions





4. How all this is related to biology ?

- Quantitative relationship between energy coupling
(H⁺ movement & ATP synthesis)
- Can I prove from the relationship that 3-4H⁺ movement
can produce 1 ATP ?



Free energy coupling

$$n [H^+]_{out} \rightleftharpoons n [H^+]_{in} \quad \dots(\Delta G_1)$$

$$ADP+Pi \rightleftharpoons ATP \quad \dots(\Delta G_2)$$

$$ADP+Pi + n [H^+]_{out} \rightleftharpoons ATP + n [H^+]_{in} \quad \dots(\Delta G_1 + \Delta G_2)$$

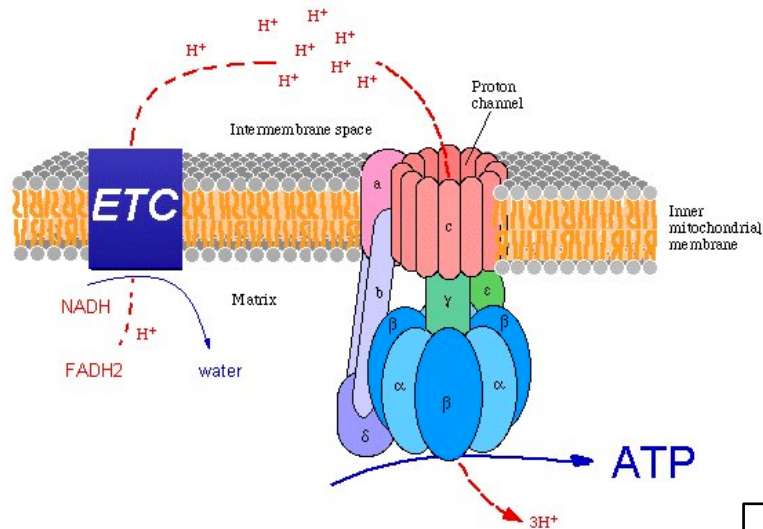
$$\Delta G_2 (ADP+Pi \rightarrow ATP) = \Delta G^0 + RT \ln ([ATP] / [ADP][Pi]) = +ve$$

$$\Delta G_1 (\text{Proton diffusion}) = \cancel{\Delta G^0} + n RT \ln ([H^+]_{in} / [H^+]_{out}) + n F \Psi = -ve$$

~~ΔG^0~~
 $= 0$

Non-Mech Work

Membrane potential
exist



Free energy coupling

$$n [H^+]_{out} \rightleftharpoons n [H^+]_{in} \quad \dots(\Delta G_1)$$



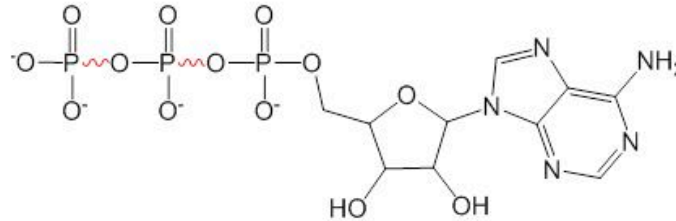
$$ADP+Pi + n [H^+]_{out} \rightleftharpoons ATP + n [H^+]_{in} \quad \dots(\Delta G_1 + \Delta G_2)$$

$$\begin{aligned} \Delta G_2 (ADP+Pi \rightarrow ATP) &= \Delta G^0 + RT \ln ([ATP] / [ADP][Pi]) \\ &= 30.5 + RT \ln(100 - 1000) = 42.4-48.3 \text{ KJ/mol} \end{aligned}$$

$$\begin{aligned} \Delta G_1 (\text{Proton diffusion}) &= \Delta G^0 + n RT \ln ([H^+]_{in} / [H^+]_{out}) + n F \psi \\ &= 0.0 + n * 2.577 \ln (1/10) - n * 96.5 * 0.1 \\ &= -46.75 \text{ (for } n=3) \\ &= -63.33 \text{ (for } n=4) \end{aligned}$$

Enough to drive ATP synthesis

ATP: the perfect energy currency for the Cell



The Nobel Prize in Chemistry 1978



"for his contribution to the understanding of biological energy transfer through the formulation of the chemiosmotic theory"



Peter D. Mitchell

The Nobel Prize in Chemistry 1997

"for their elucidation of the enzymatic mechanism underlying the synthesis of adenosine triphosphate (ATP)"



Paul D. Boyer



John E. Walker

Take Home

$$\Delta G = \Delta G^0 + RT \ln \left[\frac{\left(\frac{C_C}{C_C^0}\right)^c \left(\frac{C_D}{C_D^0}\right)^d}{\left(\frac{C_A}{C_A^0}\right)^a \left(\frac{C_B}{C_B^0}\right)^b} \right]$$

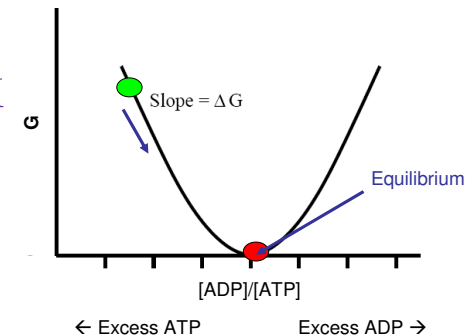
Things will stop when $\Delta G = 0$, This will fix the ratio of concentration.

$$\Delta G^0 = -RT \ln \left[\frac{\left(\frac{C_C}{C_C^0}\right)^c \left(\frac{C_D}{C_D^0}\right)^d}{\left(\frac{C_A}{C_A^0}\right)^a \left(\frac{C_B}{C_B^0}\right)^b} \right]_{\text{eqm}} = -RT \ln K_{\text{eqm}}$$

2. How much will this happen ?
3. Relation between 1 and 2 ?

➤ Displacement from equilibrium gives cells the ability to work

➤ Using free energy coupling one can prove that
3-4H⁺ movement through ATPase can produce 1 ATP.



**Best of luck for the Mid-Sem
(BT101: 26 Feb 2024, Monday)**

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